

MENORCA AP, Spain

WMO: 083140

Lat: 39.855N

Lon: 4.216E

Elev: 282

StdP: 14.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	39.3	41.5	28.5	22.7	45.1	31.7	26.4	45.5	31.7	47.7	29.1	47.8	10.0	0	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.8	89.2	71.8	87.1	71.8	85.2	71.5	76.8	83.1	75.6	82.0	74.4	80.9	9.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.0	132.5	80.2	73.5	126.1	79.5	72.2	120.5	78.7	40.4	83.3	39.2	82.1	38.1	80.7	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.5	24.2	21.2	35.1	93.0	3.0	2.1	32.9	94.6	31.1	95.8	29.4	97.0	27.2	98.6		
(5)			33.0	78.3	2.4	2.4	31.2	80.1	29.8	81.5	28.4	82.9	26.7	84.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	51.8	51.1	54.2	58.3	64.3	71.8	76.9	77.8	72.4	67.2	58.9	54.0		
	DBStd	10.27	3.97	4.40	4.05	3.72	4.09	4.29	3.39	3.43	3.91	4.36	5.11	4.37		
	HDD50	93	26	36	11	1	0	0	0	0	0	0	4	15		
	HDD65	1960	408	390	335	202	61	3	0	0	2	28	189	342		
	CDD50	4952	83	66	142	251	444	655	835	862	673	532	271	138		
	CDD65	1341	0	0	0	1	40	208	370	397	225	95	5	0		
	CDH74	9090	0	0	1	4	134	1285	3028	3337	1110	190	1	0		
	CDH80	2410	0	0	0	0	11	294	930	1015	154	6	0	0		
(14) Wind	WSAvg	9.6	10.1	10.6	10.6	10.2	9.3	8.9	8.6	8.6	8.7	9.2	10.4	9.9		
(15) Precipitation	PrecAvg	22.9	2.2	2.1	1.7	1.7	1.3	0.6	0.2	0.8	2.6	3.3	3.6	2.8		
	PrecMax	38.3	6.3	7.3	5.9	5.1	4.2	3.1	1.1	5.6	9.8	10.9	9.9	7.5		
	PrecMin	10.8	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.2		
	PrecStd	5.4	1.5	1.6	1.2	1.3	1.1	0.7	0.3	1.1	2.1	2.3	2.2	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.9	64.3	69.4	73.6	81.3	89.3	91.1	92.9	86.1	80.6	73.1	66.1		
		MCWB	57.5	56.7	58.2	59.9	65.4	68.4	71.4	72.2	73.2	70.3	65.3	60.0		
	2%	DB	61.6	61.6	65.7	70.2	77.4	86.2	88.9	89.6	83.7	78.1	70.2	64.2		
		MCWB	56.1	55.3	57.1	59.6	63.9	68.6	71.5	72.9	72.0	69.4	64.1	58.7		
	5%	DB	59.9	59.9	63.3	68.1	74.9	83.5	87.1	87.4	81.7	76.1	68.4	62.6		
		MCWB	55.1	54.3	56.0	59.0	63.5	68.3	71.5	73.1	71.1	68.5	62.8	57.9		
	10%	DB	58.6	58.2	61.1	65.8	72.4	80.8	85.2	85.3	79.7	74.1	66.6	61.0		
		MCWB	54.2	53.3	54.9	58.0	62.4	67.9	71.2	72.7	70.3	67.3	61.5	56.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.0	58.3	60.4	63.2	68.7	74.6	77.9	79.1	76.7	73.4	67.5	61.2		
		MCDB	61.7	62.1	66.4	68.9	75.4	81.5	84.9	85.6	81.3	77.0	70.5	64.1		
	2%	WB	57.3	56.8	58.5	61.4	66.7	72.5	76.0	77.3	75.1	71.4	65.7	59.9		
		MCDB	60.5	60.1	63.4	67.4	73.1	79.6	83.2	83.6	80.3	75.7	68.9	62.8		
	5%	WB	56.1	55.5	57.2	60.1	65.3	70.9	74.7	76.2	73.7	70.1	63.8	58.8		
		MCDB	59.1	58.5	61.2	65.7	71.6	78.1	81.7	82.5	79.0	74.0	67.5	61.8		
	10%	WB	54.8	54.2	55.9	58.9	63.9	69.6	73.5	75.1	72.1	68.8	62.1	57.5		
		MCDB	57.9	57.4	59.7	63.9	70.1	77.1	80.5	81.5	77.7	72.9	66.0	60.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.6	12.6	13.2	13.8	14.8	14.6	13.8	12.9	11.7	10.9	10.9		
		MCDBR	12.1	13.6	15.4	16.6	17.2	18.4	17.2	16.2	14.3	12.8	12.4	12.2		
	5% WB	MCWBR	8.9	9.6	9.3	9.0	7.7	7.4	6.8	6.7	7.0	7.1	8.5	8.8		
		MCWBR	11.0	12.4	13.3	14.4	14.5	15.4	14.6	13.4	12.4	11.3	11.5	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.348	0.385	0.407	0.422	0.423	0.438	0.428	0.417	0.394	0.363	0.336		
	taud		2.525	2.469	2.364	2.317	2.314	2.348	2.322	2.357	2.376	2.422	2.445	2.503		
	Ebn at Noon		266	276	275	276	273	272	266	266	262	255	249	253		
	Edh at Noon		24	29	36	40	41	39	40	38	35	30	26	23		
(44) All-Sky Solar Radiation	RadAvg		658	918	1367	1772	2084	2338	2303	2023	1500	1070	705	586		
	RadStd		46	103	109	105	131	100	86	87	65	85	62	41		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page